

ControlSift: Reducing Inequality in Evidence Assurance with Small-Model Triage Research

Mentor Me Collective · Google Cloud Launchpad · DeepMind AI Research Foundations
Capstone

Official UN SDG: Goal 10 – Reduced Inequalities · Dataset v1.1.0 · Protocol protocol-v1-locked

Solo project · Primary metric: macro F1 · Seed 42 · Full TEST ladder published

1. Problem statement and objectives

Organizations with limited GRC budgets face unequal access to high-quality evidence review. Template binders and policy PDFs are often treated as proof that controls operate. That false assurance harms smaller clinics, schools, nonprofits, and community providers more than enterprises that can buy deep audits.

Objective: Determine whether QLoRA adaptation of Gemma 3 1B Instruct improves five-class evidence-quality classification (SUFFICIENT, PARTIAL, INSUFFICIENT, IRRELEVANT, CONTRADICTORY) on a sealed synthetic benchmark—so triage methods can be published openly for under-resourced reviewers.

Scope: Research and education only. No production auditor. No private customer evidence. Human judgment remains authoritative.

2. Research and analysis

Problem identification used public cybersecurity and assurance sources. Modeling data is synthetic by design: 1,500 compositional evidence packets, family-isolated splits (1000/200/200/100), rule-derived labels with structural audit and scripted spot-check.

An early draft allowed TF-IDF to saturate; dataset v1.1 was hardened so lexical shortcuts no longer define a trivial ceiling.

Sealed TEST macro F1 (from docs/data/results.json): majority 0.067; TF-IDF 0.533 (challenge 0.524); Gemma zero-shot 0.080; few-shot 0.137; QLoRA 0.083 (parse success ≈ 0.435). Exact floats: 0.0667 / 0.5327 / 0.0796 / 0.1365 / 0.0827.

3. Solution development

Solution ladder: majority $\rightarrow$ TF-IDF $\rightarrow$ Gemma zero-shot $\rightarrow$ few-shot $\rightarrow$ Gemma 3 1B + QLoRA (4-bit NF4). Evaluation protocol frozen before LLM claims. Public research site includes Failure Lab, Assurance (Data Card, Model Card, Risk Register, Intended Use), and curriculum map against DeepMind AI Research Foundations.

Finding: TF-IDF leads. Few-shot is the best Gemma rung. QLoRA did not beat few-shot on TEST; low parse success makes that rung fragile. This is a sealed synthetic protocol — not a production auditor.

4. Implementation plan

Phases 0–4 complete: benchmark, seal, classical baselines, Gemma prompted baselines, QLoRA evaluation, error analysis, report/slides packaging, Pages live. Phase 5 remaining: record ≤ 5 -minute video and upload via the GCLP form with report and slides PDFs.

5. Outcomes

Sealed protocol and public receipts for all five ladder experiments.

Meaningful lexical ceiling after dataset harden (TF-IDF ~ 0.53 macro F1).

Honest negative findings: QLoRA $<$ few-shot; parse success ~ 0.435 for QLoRA.

Failure Lab + Assurance pack for responsible communication.

Remaining manual step: video recording for form upload.

6. Recommendations

Publish only from result files; keep human-in-the-loop boundaries; treat per-class weakness (especially SUFFICIENT) and parse failures as first-class findings. Do not deploy as an audit autopilot. Prefer reporting the full ladder — including when classical methods win — over claiming LLM victory.

7. Authorship attestation

This capstone is a human-directed applied research project. Assistive tools were used for drafting and engineering support; problem selection, experimental design, protocol seal, dataset decisions, and integrity rules are the author's responsibility. It is not an AI-generated project submission.